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Adaptive dynamic programming for self-triggered optimal control in industrial spatiotemporal processes

Binyan Li [a,*], Xiaopeng Cao [a], Biao Luo [b], Shan Xue [c], Jiayuan Gao [d], Shaosheng Fan [a]

[a] School of Artificial Intelligence, Changsha University of Science & Technology, Changsha, 410114, China
[b] School of Automation, Central South University, Changsha, 410083, China
[c] School of Information and Communication Engineering, Hainan University, Haikou, 570228, China
[d] School of Electrical & Information Engineering, Changsha University of Science & Technology, Changsha, 410114, China

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ABSTRACT

Industrial spatiotemporal processes are characterized by highly-dissipative partial differential equations (HPDE) with infinite dimensionality, nonlinearity, and strong spatiotemporal coupling. Precise control of such processes is critical for enhancing material quality and consistency. Since frequent updates of control inputs can easily lead to instability in state distributions, this paper investigates an adaptive self-triggered driven optimal control method for the HPDE. By dynamically predicting triggered instants, this method effectively overcomes the limitations associated with the continuous monitoring required by most existing triggered conditions. First, the self-triggered control problem of HPDE is formulated and transformed into the solution of the Hamilton-Jacobi-Bellman equation. A novel self-triggered mechanism for the HPDE system is further investigated and designed by integrating the HPDE states, control information, and Lyapunov stability analysis. Subsequently, by leveraging the formidable approximation capabilities of neural networks, adaptive dynamic programming is empowered to effectively estimate the optimal value function, thereby deriving the self-triggered optimal control. The boundedness of the self-triggered intervals and performance, and the stability of the closed-loop system are theoretically analyzed and discussed. Finally, the proposed method is applied to a typical industrial spatiotemporal process to verify its convergence, stability, and effectiveness.

1. Introduction

Industrial production processes are highly spatiotemporal in nature, which are essentially complex systems characterized by highly-dissipative partial differential equations (HPDE). Optimal control in industrial spatiotemporal processes is crucial for ensuring process stability, improving product quality, and enhancing product consistency (Zhang et al., 2018). The control methods for such systems are generally classified into two categories, reduction-based design (Christofides and Armaou, 2000; Pitchaiah & Armaou, 2010) and design-based reduction (Bensoussan et al., 2007; Fattorini, 1999). HPDE systems are inherently infinite-dimensional, nonlinear and strongly spatiotemporally coupled. These inherent characteristics inevitably result in excessively intricate system structures, thereby creating substantial difficulties in achieving accurate system modeling. This will directly lead to huge challenges in the precise control of the HPDE systems and restrict their development in the optimal control field.

Reinforcement learning (RL) can effectively solve the above problems. With in-depth research in the field of RL, adaptive dynamic programming (ADP) has gradually become representative of RL. Leveraging the optimization mechanism of dynamic programming along with RL theory, ADP establishes a synergistic dual networks framework. Specifically, an actor network is designed to approximate the optimal control law, while a critic network is employed to evaluate and approximate the optimal value function. In the complete absence of exact system dynamics, neural network (NN) approximation techniques are simultaneously employed for both the critic and actor networks to achieve optimal control. Through the alternating iterations of these two networks, continuous optimization of system performance is ultimately achieved. It is precisely the robust nonlinear approximation capability of ADP that effectively alleviates the “dimensional disaster” inherent in traditional numerical solutions. It is worth noting that implementing NN in infinite-dimensional PDE systems poses a formidable challenge regarding the construction of the activation function set. In practical ap-

* Corresponding author.
E-mail addresses: binyan.li@csust.edu.cn (B. Li), cxp@stu.csust.edu.cn (X. Cao), biao.luo@hotmail.com (B. Luo), xueshan@hainanu.edu.cn (S. Xue), 008664@csust.edu.cn (J. Gao), fss508@163.com (S. Fan).

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plications, the system's approximation accuracy improves significantly with higher time-mode dimensionality, more diverse mode combinations, and a larger activation function scale. However, this will directly lead to a significant increase in the computational cost for predicting the next triggering time and updating the weights of the neural network. Conversely, the accuracy decreases and the computational load also reduces. Striking an optimal balance between NN dimensionality, function selection, and the associated computational complexity for complex PDE systems remains a challenging task.

In the research on optimal control of ADP, Ma et al. (2026) developed an actor-critic-based optimal control scheme to guarantee prescribed tracking performance for flexible-joint robots subject to input delays. Sun and Sun (2021) proposed an alternating iterative algorithm based on the Hamilton-Jacobi-Bellman (HJB) equation by combining basis function approximation and the Newton-Leibniz formula. By combining the policy iteration algorithm with robust switching terms, the problems of the traditional ADP requiring an initial feasible control strategy and the unbounded cost function were solved in Li and Dong (2024). To address the low accuracy of ADP, which was caused by insufficient data, an alternating iteration mechanism was utilized in Ye et al. (2024) to enhance data utilization. Further, Wang et al. (2023) designed a cost function based on tracking error and its derivative, overcoming the limitation that traditional optimal control was only applicable to zero equilibrium points, and Wei et al. (2024) proposed a robust optimal parallel tracking control method, achieving uncertainty cancelation and optimal control. Recently, dynamic event-triggered ADP has seen many new applications. Peng et al. (2026) proposed an edge-based dynamic event-triggered inverse optimal formation control strategy for multiple QUAVs with attitude constraints. Qin et al. by utilizing barrier functions and boundary reconstruction techniques, dynamic event-triggered optimal safety control strategies have been developed for both multi-input mixed zero-sum games (Qin et al., 2026d) and interconnected nonlinear systems with discontinuous state constraints (Qin et al., 2026c). Furthermore, by embedding control barrier functions into nonquadratic cost functions, this event-triggered framework has been extended to achieve optimal safe tracking for interconnected systems subject to asymmetric input constraints (Qin et al., 2026a). It also includes the application of dynamic event-triggered control to human-machine cooperative systems with dynamic authority allocation (Zhang et al., 2026b), as well as the implementation of approximate optimal consensus control for unknown nonlinear multi-agent systems (Zhang et al., 2026a).

The inherent characteristics of HPDE systems in industrial production processes, lead to complex control process and large computational loads. Conventional time-triggered ADP control methods often face significant computational bottlenecks and low resource utilization efficiency, making them unsuitable for practical industrial applications. To solve the problem of optimal control in industrial production processes, it is urgent to explore a class of aperiodic optimal control methods (Heemels et al., 2012). At present, aperiodic optimal control methods are mainly divided into two types, one is the event-triggered control (Johan Åström & Bernhardsson, 1999), and the other is the self-triggered control (Velasco et al., 2026). Although they all significantly enhance the resource utilization rate of the data transmission process and save communication bandwidth through aperiodic updates, the former are often difficult to be truly applicable to actual industrial processes due to the limitation of continuous monitoring of triggered conditions.

Self-triggered control has opened up a new direction in the research of aperiodic control. By dynamically predicting the next triggered moment through the system state, control and other information at the current triggered moment, it can greatly relax the requirements for continuous monitoring of triggered conditions, further reduce the occupation of computing resources and the consumption of communication bandwidth. In recent years, it has received extensive attention from many researchers. In light of the practical requirements of industrial production, there is an urgent need to explore adaptive self-triggered control methods tailored to the specific characteristics of these industrial HPDE systems, in order to tackle the challenges associated with their control optimization.

In the research of self-triggered control methods, Yang et al. (2024) proposed a novel model predictive control strategy with self-triggered mechanism, which effectively suppressed the state fluctuations of the series hybrid power transmission system. By processing the output constraints through the delay error dependent function and the Lyapunov function, Wu et al. (2023) designed a NN adaptive self-triggered fault-tolerant controller. Aiming at the problem of communication redundancy of isolated island microgrids, a distributed self-triggered algorithm based on estimation error was proposed in Chen et al. (2023). In addition, to address the heavy computational burden in deep learning-based model predictive control for quadrotor systems, Tian et al. (2026) proposed a self-triggered tracking control framework based on the Gated Recurrent Unit neural network-ARX model, and Wu et al. (2024) utilized a self-triggered mechanism to achieve negative feedback regulation of the threshold and consensus error, thereby solving the problem of cooperative control of unmanned aerial vehicle swarfs. In addition, Hu et al. (2025b) proposed an online self-triggered critic learning framework for discrete-time nonlinear systems, which proactively determines triggering instants using current states to achieve real-time policy optimization and system stability.

However, researchers have increasingly recognized that existing optimal control methods driven by self-triggered mechanisms struggle to simultaneously meet the multifaceted demands of industrial process systems, handling complex and unknown models, ensuring precise optimal control, and achieving high resource efficiency. ADP offers an effective solution to bridge this gap. Consequently, a series of innovative studies have been conducted focusing on the deep integration of self-triggered mechanisms with ADP. For instance, for the non-zero-sum game problem of unknown nonlinear constrained systems, Liu et al. (2025) proposed an ADP algorithm that utilized the obstacle function to handle state constraints. Wei and Jiang (2025) transformed the optimal consistency tracking control problem into a multi-person zero-sum differential game problem by introducing ADP and self-triggered mechanisms. By building the nested updating policies, a self-triggered approximate optimal neural-control scheme based on ADP was proposed in Zhao et al. (2025), and Ming et al. (2025) integrated ADP with a self-triggered optimal control algorithm to address the triggered control problem in unknown nonlinear stochastic power systems. Qin et al. (2026b) utilized integral reinforcement learning and self-triggered mechanisms to propose a mixed zero-sum game optimal control framework for partially unknown nonlinear systems with mixed constraints.

In addition to the control system, the innovative neural networks and collaborative optimization strategies also demonstrate great potential. such as hierarchical adaptive attention mechanisms for industrial fault diagnosis (Wu et al., 2026), secure multidimensional anti-eavesdropping optimization in UAV-satellite communications (Liang et al., 2026), and collaborative fusion networks for real-time extreme small object detection in intelligent sports analytics (Zhang et al., 2027). These intelligent frameworks, in combination with their resource-efficient design concept, jointly promoted the development of the self-triggered ADP method for industrial processes.

Existing research indicates that while preliminary results have been achieved in the integrated application of self-triggered control and ADP, these findings are predominantly concentrated on adaptive control scenarios involving ordinary differential equation systems. From the perspective of overall field development, research of this direction is still in its infancy, and a general theoretical framework covering nonlinear, infinite-dimensional partial differential equation systems has not yet been established. Particularly, artificial intelligence has greatly enhanced the development of HPDE systems in industrial production processes at present, and there remains a distinct lack of scalable and mature methods for achieving the efficient coupling of the self-triggered mechanism and ADP technology and the stability theoretical analysis. It significantly hinders the further advancement of industrial intelli-

gent control field. This paper proposes an optimal control method for the HPDE system of industrial production processes driven by adaptive self-triggered mechanism. The main contributions of this paper are explicitly summarized as follows. 1) This paper introduces the self-triggered mechanism into the HPDE field for the first time, proposing a self-triggered ADP optimal control method suitable for industrial spatiotemporal processes, which opens up a new avenue for the efficient control of PDE systems. 2) This paper proposes a self-triggered mechanism that utilizes current state and control information to predict the next update instant, rigorously proving the existence of a strictly positive lower bound on the triggered intervals, an upper bound on the performance index, and the stability of the closed-loop system. 3) This paper designs a self-triggered ADP optimal control method based on the critic-actor network structure, and proposes an adaptive weight update strategy for the critic network that does not require initial admissible control.

The structure of the rest of this paper is as follows. Section 2 introduces the relevant knowledge of the optimal control theory for HPDE system. The construction process of the self-triggered mechanism is introduced in detail in Section 3. Section 4 presents the design process of the ADP based self-triggered control method and analyzes its stability. Section 5 provides simulation results to illustrate the effectiveness of the proposed method. Finally, a summary of this paper is provided in Section 6.

2. Problem statement

The spatiotemporal distribution characteristics exhibited by industrial production processes, can usually be described in the form of the HPDE system as shown in the following,

$$\begin{cases} \dfrac{\partial Y}{\partial t} = \mathcal{F}(\dfrac{\partial^2 Y}{\partial s^2}, \dfrac{\partial Y}{\partial s}, Y) + \bar{g}(s)u(t) \\ Y_{out}(t) = \int_{s_1}^{s_2} E(s) Y ds \end{cases} \tag{1}$$

where $Y \triangleq Y(s,t)$ represents the system state, $u(t) \in \mathbb{R}^{n_1}$ is the control input variable, $s \in [s_1, s_2]$ and $t \in [0, \infty)$ are the spatial variable and the time variable respectively. $Y_{out}(t)$ is the system output, $\bar{g}(s)$ and $E(s)$ are matrix functions of appropriate dimensions. $\mathcal{F}$ is an unknown highly dissipative differential operator. The boundary conditions and initial condition of the HPDE system are shown in (2) and (3),

$$\begin{cases} \left.\dfrac{\partial Y}{\partial s}\right|_{s=s_1} = f_1(Y,t) \\ \left.\dfrac{\partial Y}{\partial s}\right|_{s=s_2} = f_2(Y,t) \end{cases} \tag{2}$$

$$Y(s,0) = Y_0(s) \tag{3}$$

where $f_1(Y,t)$, $f_2(Y,t)$ and $Y_0(s)$ are respectively the functions that describe the boundary distribution states and the initial distribution state of the HPDE system.

Since the Fourier series can approximate any continuous function, $Y(s,t)$ can be constructed as the cumulative sum of a set of spatial basis functions and temporal modes $z_k(t)$, that is, $Y(s,t) = \sum_{k=1}^{\infty} z_k(t)\phi_k(s)$, $k = 1, \ldots, \infty$.

Define $z(t)$ as the finite-dimensional slow mode representing the dominant dynamic characteristics of the system, and $z_I(t)$ as the corresponding infinite-dimensional fast mode. According to the inner product operation of spatial basis functions, the HPDE system can be equivalently transformed into an infinite-dimensional time modes system, and its evolution process can be described as

$$\begin{cases} \dot{z}(t) = f_i(z,0) + g_i u(t) + f_i(z, z_I) - f_i(z,0) \\ \dot{z}_I(t) = \frac{1}{\chi} A_{I\chi} z_I + f_{ii}(z, z_I) + g_{ii} u(t) \\ Y_{out} = E_i z(t) + E_{ii} z_I(t) \\ z(0) = z_0, z_I(0) = z_{I0} \end{cases} \tag{4}$$

where $f_i(z, z_I) \triangleq \langle \mathcal{F}, \phi_F(s) \rangle$, $g_i \triangleq \langle \bar{B}, \phi_F(s) \rangle$, $f_{ii}(z, z_I) \triangleq \langle \mathcal{F}, \phi_I(s) \rangle - A_{I\chi} z_I / \chi$, $g_{ii} \triangleq \langle \bar{B}, \phi_I(s) \rangle$, χ is a smaller positive constant, and $A_{I\chi}$ represents an exponentially stable matrix. The corresponding function and variables in (1)–(3) are described after taking the inner product with the basis functions $\{\phi_k(s)\}_{k=1}^{\infty}$, as shown in the following

$$\begin{cases} z(t) = \langle Y(s,t), \phi_F(s) \rangle \triangleq \left[z_1(t), \cdots, z_N(t)\right]^T, \phi_F(s) \triangleq \left[\phi_1(t), \cdots, \phi_N(t)\right]^T \\ z_I(t) = \langle Y(s,t), \phi_I(s) \rangle \triangleq \left[z_{N+1}(t), \cdots, z_\infty(t)\right]^T, \phi_I(s) \triangleq \left[\phi_{N+1}(t), \cdots, \phi_\infty(t)\right]^T \\ E_i \triangleq \int_{s_1}^{s_2} E(s)\phi_F^T(s) ds, E_{ii} \triangleq \int_{s_1}^{s_2} E(s)\phi_I^T(s) ds \\ z_0(t) = \langle Y_0(s), \phi_F(s) \rangle, z_{I0}(t) = \langle Y_0(s), \phi_I(s) \rangle \end{cases} \tag{5}$$

By combining the Karhunen–Loève decomposition (KLD) method with the singular perturbation theory (Christofides & Daoutidis, 1997), a finite-dimensional low-order slow subsystem can be derived based on (4), as presented in the equation below,

$$\begin{cases} \dot{z}(t) = f_i(z(t)) + g_i u(t), z(0) = z_0 \\ Y_{out\,F}(t) = E_i z(t) \end{cases} \tag{6}$$

The objective of conducting the optimal control research on the above-mentioned HPDE system is reflected in making the HPDE system (1) stable by finding a feedback control strategy $u(t)$, where it updates periodically under time-triggered conditions. Based on the system output and control information, the performance index $J_u(Y_0(s))$ related to space in the infinite time domain is constructed, as expressed in the following,

$$J_u(Y_0(s)) = \int_0^{+\infty} (\left\|Y_{out}(t)\right\|^2 + \|u(t)\|_R^2) dt \tag{7}$$

where R is the control weight matrix. Accordingly, the optimal control problem of the HPDE system (1) can be transformed into the control problem of the low-order slow subsystem (6) through (4)–(7). From (6), $J_u(Y_0(s))$ can be further constructed,

$$J_u(z_0) = \int_0^{+\infty} (\|z(t)\|_Q^2 + \|u(t)\|_R^2) dt \tag{8}$$

Combining (7)–(8), it can be further rewritten as (Luo et al., 2014),

$$J_u(Y_0(s)) = J_u(z_0) + J_{ii}(z, z_I) \tag{9}$$

Next, construct the Hamiltonian function $H(z, u, \nabla J_u)$ for the HPDE system (1) derived from (6) and (8), that is,

$$H(z, u, \nabla J_u) = \nabla J^T (f_i(z) + g_i u) + \|z\|_Q^2 + \|u\|_R^2 \tag{10}$$

On this basis, the optimal control problem of the system (1) studied in this paper can be converted into the problem of solving the HJB equation,

$$\min_u H(z, u(z), \nabla J^*(z)) = 0 \tag{11}$$

where $J^*(z)$ is the optimal value function, satisfying $J^*(z) \geq 0$ and $J^*(0) = 0$.

Thus, based on (11) and the principle of first-order optimality (Lewis et al., 2012), the optimal control strategy $u^*(z)$ of the HPDE system (1) is derived, and its expression is

$$u^*(z) = -\frac{1}{2} R^{-1} g_i^T \nabla J^*(z) \tag{12}$$

In view of this, substituting (12) into (11), the HJB equation can be further transformed into the following form,

$$\|z\|_Q^2 - \frac{1}{4} \left[\nabla J^*(z)\right]^T g_i R^{-1} g_i^T \nabla J^*(z) + \left[\nabla J^*(z)\right]^T f_i(z) = 0 \tag{13}$$

3. Self-triggered mechanism

Self-triggered control has the feature of avoiding continuous periodic monitoring of triggered conditions, which can further reduce the consumption of computing resources and communication bandwidth in the control process of the HPDE system.

In order to achieve the self-triggered optimal control of the studied system, this section proposes a self-triggered mechanism design method for HPDE system. On account of the problem statement above, one can further present the self-triggered control problem. Furthermore, by integrating system stability analysis method, we rationally construct and design a self-triggered mechanism. Moreover, theoretical analysis and discussion are conducted regarding the boundedness of the triggered intervals and performance index under the proposed mechanism.

3.1. Control problem

Leveraging the current system state and control information, the self-triggered mechanism proactively predicts the exact time of the next control update. During the predicted interval, no state monitoring or condition checks are required,NE thereby significantly reducing computational overhead.

To this end, it is imperative to define the available state of the self-triggered controller and the modal error, and to construct both the self-triggered intervals and the control strategy.

Since the self-triggered control updating are only performed at triggered moments, it can be constructed as follows, $\mu(t) \triangleq u(t), t = t_j$, $\mu(t) \triangleq u(t_j), t \in (t_j, t_{j+1})$, where both t_j and t_{j+1} represent the triggered moments, and j is a positive integer. The available states of the self-triggered controller are defined as $\bar{Y}(s,t) \triangleq Y(s,t), t = t_j$ and $\bar{Y}(s,t) \triangleq Y(s,t_j), t \in (t_j, t_{j+1})$.

Due to self-triggered update feature, there is a difference between the state held by the controller and the actual system state. Therefore, it is necessary to construct the dominant dynamic time modal error $e(t)$. It is defined as the error between the current time mode held by the self-triggered controller and the mode of the HPDE system at the current moment, that is,

$$e(t) \triangleq \bar{z}(t) - z(t) \tag{14}$$

where $\bar{z}(t) \triangleq z(t)$ with $t = t_j$, and $\bar{z}(t) \triangleq z(t_j)$ with $t \in (t_j, t_{j+1})$. Furthermore, the time interval between two adjacent self-triggered moments is defined as the triggered interval $\varpi(z(t))$, and its expression is

$$\varpi(z(t)) \triangleq t_{j+1} - t_j \tag{15}$$

The optimal control problem under the self-triggered mechanism stems from the corresponding problem with time-triggered mechanism. This indicates that for the studied HPDE system (1), according to the optimal control strategy $u^*(z)$ (12), the self-triggered optimal control strategy $\mu^*(\bar{z}(t))$ is

$$\mu^*(\bar{z}(t)) \triangleq u^*(\bar{z}(t)) = -\frac{1}{2} R^{-1} g_i^T \nabla J^*(\bar{z}) \tag{16}$$

By substituting (16) into (1), the closed-loop HPDE system under self-triggered control can be obtained and shown below,

$$\begin{cases} \dfrac{\partial Y}{\partial t} = \mathcal{F}(\dfrac{\partial^2 Y}{\partial s^2}, \dfrac{\partial Y}{\partial s}, Y) + \bar{g}(s)\mu^*(\bar{z}) \\ Y_{out}(t) = \int_{s_1}^{s_2} E(s) Y \, ds \end{cases} \tag{17}$$

3.2. Self-triggered mechanism

The implementation of self-triggered optimal control relies on the dynamic prediction of the triggered moments. Designing an appropriate self-triggered mechanism is essential (Hu et al., 2025a). For this purpose, this section relies on the idea of aperiodic updating of event-triggered mechanism. By combining the Lyapunov stability analysis theory, the reverse Lyapunov theory, and the comparison lemma (Khalil & Grizzle, 2002), we can construct and design the triggered conditions and triggered moments of the HPDE system. This section proposes the design method of a self-triggered mechanism, and gives some reasonable assumptions as shown below.

Assumption 1. If the optimal control strategy $u^*(z)$ (12) satisfies Lipschitz continuity, there exists a constant λ_u that satisfies the following inequality relation $\|u^*(z_1) - u^*(z_2)\| \le \lambda_u \|z_1 - z_2\|$ with $z_1, z_2 \in \mathcal{X}$.

Assumption 2. For the function $f_i(z)$ shown in (6), it possesses Lipschitz continuity, that is, $\|f_i(z)\| \le \lambda_f \|z\|$ holds with $\lambda_f > 0$.

Assumption 3. Suppose λ_1 and λ_2 represent the minimum and maximum singular values of the matrix Q respectively, and it has $\lambda_1 \|z\|^2 \le \|z\|_Q^2 \le \lambda_2 \|z\|^2$. If the strategy $u^*(z)$ is Lipschitz continuous, there exist $\lambda_3, \lambda_4 > 0$ to ensure $\lambda_3 \|z\| \le \|u^*(z)\| \le \lambda_4 \|z\|$ and $\lambda_3 \|z\| \le \|\mu^*(\bar{z})\| \le \lambda_4 \|z\|$ hold. Moreover, it yields $\|\nabla J^*(z)\| - \|\nabla J^*(\bar{z})\| \le \lambda_V \|e\|$ with $\lambda_V > 0$ for the condition $\|\nabla J^*(\bar{z})\| < \|\nabla J^*(z)\|$.

Assumption 4. For the HPDE system (1), it can be known from (7) to (9) that there exists a constant β_{ii} such that the relation $J_{ii}(z, z_I) \le \beta_{ii} J_u(z_0)$ is satisfied among the functions $J_u(Y_0(s))$, $J_u(z_0)$ and $J_{ii}(z, z_I)$.

Assumption 5. Assume that the system states operate within a predefined bounded compact set $\Omega \in \mathbb{R}^n$. Within this compact set, the activation function $\eta(z)$, the weight ω and the approximation error ζ_i are all bounded (Luo et al., 2014), and there exist constants Ω_η, Ω_ω and Ω_{ζ_i} such that $\|\eta(z)\| \le \Omega_\eta$, $\|\omega\| \le \Omega_\omega$ and $\|\zeta_i\| \le \Omega_{\zeta_i}$ hold true. Meanwhile, the gradients of the approximation error $\nabla\zeta_i(z)$ and the activation function $\nabla\eta(z)$ satisfy the following inequality relation $\|\nabla\zeta_i(z)\| \le \Omega_{G\zeta_i}$ and $\|\nabla\eta(z)\| \le \Omega_{G\eta}$, where $\Omega_{G\zeta_i}$ and $\Omega_{G\eta}$ are positive constants.

It should be emphasized that Assumption 5 precisely defines the boundary of the existence of ideal NN parameters within the compact set. During the online learning process, the boundedness of activation functions, weights, and approximation errors is strictly guaranteed by the proposed adaptive weight update strategy and the subsequent SGUUB stability analysis.

Definition 1. Consider the closed-loop system (17) states z(t) and the initial time $t_0 \ge 0$. If for any compact set Ω and all initial conditions $z(t_0) \in \Omega$, there exists a constant P and a control time $T = T(P, z(t_0)) \ge 0$ such that the system satisfies $\|z(t)\| \le P$, $\forall t \ge t_0 + T$, then this system is called semi-globally uniformly ultimately bounded (SGUUB).

Remark 1. The assumptions in this paper are standard, well-founded, and essential for the theoretical analysis of HPDE systems. Specifically, Assumptions 1, 2 and 3 establish the Lipschitz continuity and inherent boundedness of the system dynamics $f_i(z)$, control variables, and performance index gradients. Assumption 4 defines the specific boundary conditions for the performance indices J_{ii} and J_u. Assumption 5 guarantees the boundedness of the ideal neural network parameters and approximation errors. Supported by extensive existing research (Hu et al., 2025a; Liu et al., 2025; Ming et al., 2023, 2025; Qin et al., 2026a,b,c,d; Wei & Jiang, 2025; Zhang et al., 2026a,b; Zhao et al., 2025), these assumptions are mathematically reasonable and universally applicable.

Theorem 1. *Consider the closed-loop HPDE system (17), and Assumptions 1 to 3 hold. Under the action of the self-triggered control strategy (16), if the triggered condition is constructed as*

$$\begin{aligned} \|e(t)\|^2 \le & \frac{\Gamma_1 \left\| \langle Y(s,t), \phi_F(s) \rangle \right\|_Q^2 + \Gamma_2 \, \| \, \mu^*(\bar{z}) \, \|_R^2}{\lambda_u^2 \|r\|^2 + \Gamma_3} \\ = & \frac{\Gamma_1 \|\bar{z}\|_Q^2 + \Gamma_2 \, \| \, \mu^*(\bar{z}) \, \|_R^2}{\lambda_u^2 \|r\|^2 + \Gamma_3} = e_T^2 \end{aligned} \tag{18}$$

and is violated simultaneously, the next self-triggered moment t_{j+1} can be calculated and articulated in the following,

$$t_{j+1} = t_j + \frac{1}{d} \ln(1 + (\frac{d}{a\|\bar{z}\| + c} e_T)) \tag{19}$$

where a, b, c, α and θ are adjustable parameters. Then, the closed-loop HPDE system under the self-triggered control maintains SGUUB stability.

Proof. See the Appendix A. □

3.3. Boundedness analysis

Upon the designed self-triggered mechanism, this subsection further analyzes the boundedness of the self-triggered intervals. The boundedness plays a pivotal role in determining whether Zeno behavior can be effectively precluded under the designed self-triggered mechanism (Ming et al., 2023). Meanwhile, the boundedness of the performance index of the HPDE system can also be guaranteed according to the designed self-triggered mechanism (18) and (19), which is also the difference from the existing research (Yang & Wei, 2021).

Lemma 1. *For the HPDE system with the self-triggered control (17), the corresponding self-triggered interval $\varpi(z(t))$ is defined as shown in (15). Under the self-triggered mechanism (18) and (19), a uniform positive lower bound $\varpi_{\min}(z(t)) > 0$ exists for all initial conditions within the compact operating region and bounded disturbances, thereby strictly avoiding Zeno behavior (Yang & He, 2021).*

As demonstrated in the proof of Theorem 1, the closed-loop system is SGUUB. This guarantees that for any bounded initial conditions and disturbances, all system states are strictly confined within a compact set. Consequently, a uniform positive lower bound for the triggered interval is inherently ensured, strictly avoiding Zeno behavior as mathematically verified in (A.17).

Theorem 2. *Consider the closed-loop HPDE system (17) and the performance index $J_\mu(Y_0(s))$ (9). If there exists a constant β_{ii} such that Assumption 4 holds true, then under the effect of the self-triggered control strategy (16) and the self-triggered mechanism (18) and (19), the upper bound of $J_\mu(Y_0(s))$ satisfies $J_\mu(Y_0(s)) \le \Omega_p$, where $\Omega_p \triangleq (1+\beta_{ii})J_{u^*}(z_0)/\prod_2$, and $\prod_2 \triangleq (1-\alpha)(1-\theta)$.*

Proof. This proof process consists of two parts.

1) Compared with low-order slow subsystem under periodic control, we prove that there is an upper bound for a performance index in the self-triggered control. 2) Compared with low-order slow subsystem under self-triggered, we prove that there is an upper bound for a performance index in the HPDE system under self-triggered. First, based on (A.3), the relationship below is obtained,

$$\|\bar{z}(t)\|_Q^2 + \|\mu^*(\bar{z})\|_R^2 \le -\frac{1}{\prod_2}\dot{J}^*(z) \tag{20}$$

From (20), it is apparent that this part only considers the low-order slow subsystem under self-triggered control, that is, it satisfies $z_1(t) = 0$. Then, one can conclude that $\dot{J}^*(z) = [\nabla J^*(z)]^T(f_i(z,0) + g_i\mu^*(\bar{z})) \le 0$ holds. Meanwhile $\lim_{t\to\infty} z_1(t) = 0$ holds for low-order slow subsystems. Thus, derived from (8) and (20), we have

$$\begin{aligned} J_\mu(z_0) &= \int_0^{+\infty} (\|\bar{z}(t)\|_Q^2 + \|\mu^*(\bar{z})\|_R^2)dt \\ &\le -\frac{1}{\prod_2}\int_0^{+\infty} d(J^*(z)) \\ &= \frac{1}{\prod_2}J_{u^*}(z_0) \end{aligned} \tag{21}$$

Leveraging (8), (9) and Assumption 4, one yields the relation,

$$\begin{aligned} J_\mu(Y_0(s)) &= \int_0^{+\infty} (\|Y_{out}(t)\|^2 + \| \mu^*(\bar{z})\|_R^2)dt \\ &\le (1+\beta_{ii})\int_0^{+\infty} (\|\bar{z}(t)\|_Q^2 + \|\mu^*(\bar{z})\|_R^2)dt \\ &= (1+\beta_{ii})J_\mu(z_0) \end{aligned} \tag{22}$$

Combining (21) and (22) above, the subsequent expression can be obtained,

$$J_\mu(Y_0(s)) \le \frac{(1+\beta_{ii})}{\prod_2}J_{u^*}(z_0) \triangleq \Omega_p \tag{23}$$

In conclusion, an upper bound on the performance index of the HPDE system under self-triggered optimal control is established. This concludes the proof. □

In practical applications, under the self-triggered mechanism, the controller only utilizes the state information at discrete triggering moments, while the model reduction process ignores the influence of infinite-dimensional fast modes. Therefore, Theorem 2 provides a rigorous mathematical framework for precisely answering the extent of actual performance deterioration under these two challenges.

Remark 2. In practical applications, if parameters $\prod_2$ are drastically adjusted to significantly reduce communication and computing resources, the system's performance will inevitably suffer. Mathematically, this manifests as a sharp increase in the upper bound of performance indicators, denoted as Ω_p. On the contrary, if better performance is required, it will inevitably lead to an increase in trigger frequency and excessive consumption of resources. Therefore, Theorem 2 provides engineers with a quantitative mathematical boundary, helping them find the optimal balance point between acceptable performance degradation and the desired resource savings.

4. Self-triggered approximate neural controller

This section proposes a self-triggered optimal control method for HPDE system based on ADP, which integrates NN technology to approximate the value function. In approximating the optimal value function for HPDE system via NN, the formulation and scale of the activation function set usually depend heavily on the identified finite-dimensional slow modes and physical prior knowledge. This inherently introduces a difficult trade-off as selecting too many neurons leads to excessive computational complexity, while selecting too few results in unacceptable approximation errors. Regarding the HPDE system in this paper, an in-depth analysis of its mode characteristics, combined with a physical understanding of the process, enables the development of an activation function set. This set strikes an optimal balance between approximation fidelity and efficiency. This method tackles the challenge of designing the optimal controller when the analytical solution to the HJB equation is difficult to obtain accurately. We begin this section by designing a self-triggered controller grounded in the critic-actor networks, and then proceed to analyze and discuss the stability of the resulting closed-loop HPDE system.

Within the critic network, the optimal value function is approximated by an appropriately designed NN. The value function $J^*(z)$ of the HJB equation can be expressed as $J^*(z) = [\eta(z)]^T\omega^* + \zeta_i(z)$, where $\eta(z) \in \mathbb{R}^{n_2}$ and $\omega^* \in \mathbb{R}^{n_2}$ are the activation function and the weight vector, n_2 refers to the number of hidden neurons, and $\zeta_i(z)$ is the approximate error of NN. By taking the partial derivative of the valued function $J^*(z)$ with respect to z, $\nabla J^*(z) = [\nabla\eta(z)]^T\omega^* + \nabla\zeta_i(z)$ can be obtained, where $\nabla J^*(z)$ is the gradient of $J^*(z)$, $\nabla\eta(z)$ and $\nabla\zeta_i(z)$ are the gradients of $\eta(z)$ and $\zeta_i(z)$ respectively.

However, determining the ideal weight ω^* of an NN accurately still remains a significant challenge. Consequently, the critic network provides an approximate representation of the optimal value function $J^*(z)$, from which it follows

$$\hat{J}(z) = [\eta(z)]^T\hat{\omega} \tag{24}$$

$$\nabla\hat{J}(z) = [\nabla\eta(z)]^T\hat{\omega} \tag{25}$$

where $\hat{\omega}$ is the estimated weight value, and $\nabla\hat{J}(z)$ is the gradient of $\hat{J}(z)$.

For the actor network, the self-triggered control strategy $\boldsymbol{\mu}(\bar{z})$ can be described by using (16) and (25), as presented in the equation below,

$$\boldsymbol{\mu}(\bar{z}) = -\frac{1}{2}R^{-1}g_i^T[\nabla\eta(\bar{z})]^T\omega \tag{26}$$

where the weight ω satisfies

$$\omega(t) = \begin{cases} \hat{\omega}(t) & t = t_j \\ \hat{\omega}(t_j) & t \in (t_j, t_{j+1}) \end{cases}$$

According to (10), (25), and (26), the HJB equation can be rewritten as

$$\begin{aligned} &H(z,\mu(\bar{z}),\nabla\hat{J}(z)) \\ &=[\nabla\hat{J}(z)]^T[f_i(z(t))+g_i\mu(\bar{z})]+\|z(t)\|_R^2+\|\mu(\bar{z})\|_R^2 \\ &=\hat{\omega}^T\nabla\eta(z)[f_i(z(t))+g_i\mu(\bar{z})]+\|z(t)\|_R^2+\|\mu(\bar{z})\|_R^2 \\ &\triangleq\zeta_{ii}(z,\hat{\omega}) \end{aligned} \tag{27}$$

where ζ_{ii} is called the residual. Based on this, the square of the residual of the HJB equation is constructed according to (27), i.e., $\gamma(z,\hat{\omega})\triangleq(1/2)\zeta_{ii}^2(z,\hat{\omega})$. The true weights of NN can be approximated by minimizing $\gamma(z,\hat{\omega})$.

To achieve control optimization without requiring an initial admissible control, we employ the following adaptive weight update law for the critic network. Structurally, the proposed design comprises a two-tier logic, centered on the adaptive switching parameter ψ_2,

$$\begin{cases}\psi_2=0, & \text{when} \quad [\nabla\bar{J}]^T[f_i(z)+g_i\mu(\bar{z})]\le 0\\ \psi_2=0.5, & \text{else.}\end{cases} \tag{28}$$

Further, the adaptive weight update strategy $\dot{\hat{\omega}}$ of the critic network is designed in combination with the gradient descent method, as illustrated in the following,

$$\dot{\hat{\omega}}=-\underbrace{\frac{\psi_1\delta\zeta_{ii}(z,\hat{\omega})}{\left(\delta^T\delta+1\right)^2}}_{\Lambda_1}+\underbrace{\psi_2\nabla\eta(z)\Omega\nabla\bar{J}}_{\Lambda_2} \tag{29}$$

where $\delta\triangleq\nabla\eta[f_i(z)+g_i\mu(\bar{z})]$, $\psi_1>0$ represents the learning rate, and $\Omega=g_iR^{-1}g_i^T$. The radially unbounded Lyapunov function $\bar{J}$ is selected as $\bar{J}(z)=\frac{1}{2}z^TKz$ and satisfies $[\nabla\bar{J}]^T[f_i(z)+g_iu^*(z)]\le 0$, where K is a symmetric positive-definite matrix. The adaptive weight update strategy $\dot{\hat{\omega}}$ of critic network, as designed above, comprises two primary components. The first component is a normalized gradient descent term based on residual minimization Λ_1, which drives the weights to update along the gradient direction that reduces the ζ_{ii}. The second component is an auxiliary term Λ_2 introduced in this paper, whose activation is strictly governed by the parameter ψ_2 in (28). When the initially provided control input fails to stabilize the system, this auxiliary term intervenes to steer the weight update direction, which in turn adjusts the self-triggered control strategy to force the system states back onto a stabilizing trajectory.

There exist constants $\bar{\Omega}_i$, $\underline{\Omega}_\delta$, $\bar{\Omega}_\delta$ and $\bar{\Omega}_{G\bar{J}}$ that make the following relation hold,

$$\begin{cases}\|f_i(z)\|\le\bar{\Omega}_i\\ \underline{\Omega}_\delta\le\|\delta\|\le\bar{\Omega}_\delta\\ \|\nabla\bar{J}(z)\|\le\Omega_{G\bar{J}}\end{cases} \tag{30}$$

Meanwhile, the weight error $\tilde{\omega}(t)$ is defined as the difference between the estimated weight $\hat{\omega}$ and the ideal value ω^*, that is, $\tilde{\omega}(t)\triangleq\hat{\omega}(t)-\omega^*(t)$. It further follows that the relationship $\dot{\tilde{\omega}}(t)=\dot{\hat{\omega}}(t)$ between $\dot{\tilde{\omega}}(t)$ and $\dot{\hat{\omega}}(t)$ is satisfied. Correspondingly, define the control error $\tilde{\mu}(\bar{z})$ as $\tilde{\mu}(\bar{z})\triangleq\mu(\bar{z})-\mu^*(\bar{z})$, the estimation error of the value function $\tilde{J}(z)$ as $\tilde{J}(z)\triangleq\hat{J}(z)-J^*(z)$. Combining (16), Assumption 5 and the weight error $\tilde{\omega}(t)$, the following relationship can be derived,

$$\begin{cases}J^*(z)=[\eta(z)]^T(\hat{\omega}-\tilde{\omega})+\zeta_i(z)\\ \tilde{J}(z)=[\eta(z)]^T\tilde{\omega}-\zeta_i(z)\\ \|\nabla J^*(z)\|\le\Omega_{G\eta}\Omega_\omega+\Omega_{G\zeta_i}=\Omega_{GJ}\\ \|\mu^*\|=\left\|-\frac{1}{2}R^{-1}g_i^T\nabla J^*\right\|\le\Omega_\mu\end{cases} \tag{31}$$

Theorem 3. *For the HPDE system (17), the adaptive weight update strategy $\dot{\hat{\omega}}$ is shown in (29), the self-triggered control strategy $\boldsymbol{\mu}(\bar{z})$ is given by (26), and both Assumptions 1 and 3–5 are true. Then, the self-triggered mechanism is as follows,*

$$t_{j+1}=t_j+\frac{1}{d}\ln\left(1+\left(\frac{d}{a\|\bar{z}\|+c}\times\sqrt{\frac{\Gamma_1\|\bar{z}\|_Q^2+\Gamma_2\parallel\mu(\bar{z})\parallel_R^2}{\lambda_u^2\parallel r\parallel^2+\Gamma_3}}\right)\right) \tag{32}$$

Governed by the above self-triggered mechanism and control strategy, the closed-loop HPDE system is guaranteed to be SGUUB under the $\mathcal{L}_2$ norm.

Proof. Before proving the stability of the HPDE system (17) with the self-triggered mechanism, we select a smooth Lyapunov candidate function $\mathcal{L}(t)$ initially, as demonstrated below,

$$\mathcal{L}(t)=\frac{1}{2}\tilde{\omega}^T\tilde{\omega}+\bar{J}(z)+J^*(z)+J_i(z_I) \tag{33}$$

Part One: $t\in[t_j,t_{j+1})$, the self-triggered condition is satisfied, the HPDE system (17) is not required to update its self-triggered control. The time derivative of the first term in (33) can be expressed as

$$\begin{aligned}\frac{d(\frac{1}{2}\tilde{\omega}^T\tilde{\omega})}{dt}&=\tilde{\omega}^T[-\frac{\psi_1\delta\zeta_{ii}(z,\hat{\omega})}{(\delta^T\delta+1)^2}+\psi_2\nabla\eta(z)\Omega\nabla\bar{J}] \\ &=\underbrace{-\frac{\psi_1\zeta_{ii}(z,\hat{\omega})\tilde{\omega}^T\delta}{(\delta^T\delta+1)^2}}_{\mathcal{A}_1}+\underbrace{\psi_2\tilde{\omega}^T\nabla\eta(z)g_iR^{-1}g_i^T\nabla\bar{J}}_{\mathcal{A}_2}\end{aligned} \tag{34}$$

Based on Assumptions 1 and 3, by substituting (27) and (29), the first term of (34) can be derived,

$$\begin{aligned}\mathcal{A}_1&=-\psi_1\tilde{\omega}^T\frac{\delta^T\delta\hat{\omega}+[\|z(t)\|_Q^2+\|\mu(\bar{z})\|_R^2]\delta}{\left(\delta^T\delta+1\right)^2} \\ &=-\psi_1\frac{\delta^T\delta\tilde{\omega}^T(\tilde{\omega}+\omega^*)}{\left(\delta^T\delta+1\right)^2}-\psi_1\frac{[\|z(t)\|_Q^2+\|\mu(\bar{z})\|_R^2]\tilde{\omega}^T\delta}{\left(\delta^T\delta+1\right)^2} \\ &\le-\frac{\psi_1\underline{\Omega}_\delta^2\|\tilde{\omega}\|(\|\tilde{\omega}\|+\Omega_\omega)}{\left(1+\bar{\Omega}_\delta^2\right)^2}-\frac{\psi_1\underline{\Omega}_\delta(\lambda_1+\lambda_3^2\Omega_R)\|z\|^2\|\tilde{\omega}\|}{\left(1+\bar{\Omega}_\delta^2\right)^2} \\ &=-\psi_1\hbar_{F1}\|\tilde{\omega}\|^2-\psi_1\hbar_{F2}\|\tilde{\omega}\|-\psi_1\hbar_{F3}\|z\|^2\|\tilde{\omega}\|\end{aligned} \tag{35}$$

where $\hbar_{F1}\triangleq\underline{\Omega}_\delta^2/\left(1+\bar{\Omega}_\delta^2\right)^2$, $\hbar_{F2}\triangleq\underline{\Omega}_\delta^2\Omega_\omega/\left(1+\bar{\Omega}_\delta^2\right)^2$, and $\hbar_{F3}\triangleq\underline{\Omega}_\delta^2(\lambda_1+\lambda_3^2\Omega_R)/\left(1+\bar{\Omega}_\delta^2\right)^2$.

The time derivative of $\bar{J}(z)$ in (33), is given by

$$\dot{\bar{J}}(z)=[\nabla\bar{J}(z)]^T(f_i(z)+g_i\mu(\bar{z})) \tag{36}$$

Building on the relationship between $[\nabla\bar{J}(z)]^T(f_i(z)+g_i\mu(\bar{z}))$ and ψ_2 discussed earlier, we analyze (36) and the second term of (34) together, the analysis covers two cases.

1) If $[\nabla\bar{J}(z)]^T(f_i(z)+g_i\mu(\bar{z}))\le 0$, then $\dot{\bar{J}}(z)+\mathcal{A}_2\le 0$ can be derived.

2) Conversely, further analysis is required in combination with mathematical transformation. From (26), Assumptions 1, 3, 4, and the corresponding numerical transformation yields the following inequality,

$$\begin{aligned}\dot{\bar{J}}(z)+\mathcal{A}_2&=[\nabla\bar{J}(z)]^T(f_i(z)+g_i\mu^*(z))+[\nabla\bar{J}(z)]^Tg_i(\mu(\bar{z}) \\ &\quad-\mu^*(z))+\frac{1}{2}\tilde{\omega}^T\nabla\eta(z)g_iR^{-1}g_i^T\nabla\bar{J}(z) \\ &\le\|\nabla\bar{J}(z)\|\|\mu(\bar{z})-\mu^*(z)\|+\frac{1}{2}\left\|\tilde{\omega}^T\right\|\|\nabla\eta(z)\|\left\|g_ig_i^T\right\|\left\|R^{-1}\right\|\|\nabla\bar{J}(z)\| \\ &\le\Omega_{G\bar{J}}(\lambda_4\|z\|+\Omega_\mu)+\frac{1}{2}\|\tilde{\omega}\|\Omega_{G\eta}\Omega_g^2\Omega_R^{-1}\Omega_{G\bar{J}} \\ &\le\hbar_{F4}\|z\|+\hbar_{F5}\|\tilde{\omega}\|+\hbar_{F6}\end{aligned} \tag{37}$$

where $\hbar_{F4}\triangleq\Omega_{G\bar{J}}\lambda_4$, $\hbar_{F5}\triangleq\frac{1}{2}\Omega_{G\eta}\Omega_g^2\Omega_R^{-1}\Omega_{G\bar{J}}$, and $\hbar_{F6}\triangleq\Omega_{G\bar{J}}\Omega_\mu$.

Taking the time derivative of $J^*(z)$ along with (31), (A.2), (A.4), and Assumptions 1, 3, we notice that

$$\begin{aligned}\dot{J}^*(z)&=[\nabla J^*(z)]^T(f_i(z,0)+g_i\mu(\bar{z})+f_i(z,z_I)-f_i(z,0)) \\ &=\left\|r^Tu^*(z)-r^T\mu(\bar{z})\right\|^2-\left\|r^T\mu(\bar{z})\right\|^2-[\bar{z}^TQ\bar{z}-2\bar{z}Qe+e^TQe] \\ &\quad+[\nabla J^*(z)]^T(f_i(z,z_I)-f_i(z,0)) \\ &\le\lambda_u^2\|r\|^2\|e(t)\|^2-\left\|\mu^*(\bar{z})\right\|_R^2-(1-\alpha)(1-\theta)\|\bar{z}\|_Q^2 \\ &\quad-(1-\alpha)\theta\|\bar{z}\|_Q^2+(\frac{1}{\alpha}-1)\lambda_{\max}(Q)\|e(t)\|^2+\Omega_{GJ}\Upsilon_{a3}\|z_I\|_{\mathbb{R}^\infty} \\ &\le-(1-\alpha)(1-\theta)\|\bar{z}\|_Q^2-\frac{1}{4}(1-\alpha)(1-\theta)\end{aligned}$$

$$\times \underline{\sigma}(g_i R^{-1} g_i^T)\|\nabla J^*(\bar{z})\|^2 + \Omega_{GJ}\Upsilon_{a3}\|z_I\|_{\mathbb{R}^\infty}$$
$$= -\prod_2\|\bar{z}\|^2 + \hbar_{F7}\|z_I\|_{\mathbb{R}^\infty} - \frac{1}{4}\prod_1\prod_2\|\nabla J^*(\bar{z})\|^2 \tag{38}$$

where $\hbar_{F7} \triangleq \Omega_{GJ}\Upsilon_{a3}$.

By considering (A.1), (A.4), (26), and (31), the time derivative of $J_i(z_I)$ becomes

$$\begin{aligned}\dot{J}_i(z_I) &= \frac{1}{\chi}[\nabla J_i(z_I)]^T A_{I\chi} z_I + [\nabla J_i(z_I)]^T f_{ii}(z, z_I) + [\nabla J_i(z_I)]^T g_{ii}(\mu^*(\bar{z}) + \mu(\bar{z}))\\ &= \frac{1}{\chi}[\nabla J_i(z_I)]^T A_{I\chi} z_I + [\nabla J_i(z_I)]^T f_{ii}(z, z_I) + [\nabla J_i(z_I)]^T g_{ii}\mu^*(\bar{z})\\ &- \frac{1}{2}[\nabla J_i(z_I)]^T g_{ii} R^{-1} g_{ii}^T [\nabla\eta(\bar{z})]^T\tilde{\omega} + \frac{1}{2}[\nabla J_i(z_I)]^T g_{ii} R^{-1} g_{ii}^T \nabla\zeta_i(\bar{z})\\ &\le -\frac{\Upsilon_{b3}}{\chi}\|z_I\|_{\mathbb{R}^\infty}^2 + (\Upsilon_{a1}\|z\| + \Upsilon_{a2}\|a_I\|_{\mathbb{R}^\infty})\Upsilon_{b4}\|z_I\|_{\mathbb{R}^\infty} + \Upsilon_{b4}\Upsilon_{b5}\Omega_\mu\|z_I\|_{\mathbb{R}^\infty}\\ &+ \frac{1}{2}\Upsilon_{b4}\Upsilon_{b5}\Omega_R^{-1}\Omega_g\Omega_{G\eta}\|z_I\|_{\mathbb{R}^\infty}\|\tilde{\omega}\| + \frac{1}{2}\Upsilon_{b4}\Upsilon_{b5}\Omega_R^{-1}\Omega_g\Omega_{G\zeta_i}\|z_I\|_{\mathbb{R}^\infty}\\ &= (\hbar_{I1} - \frac{\Upsilon_{b3}}{\chi})\|z_I\|_{\mathbb{R}^\infty}^2 + \hbar_{I2}\|z\|\|z_I\|_{\mathbb{R}^\infty} + \hbar_{I3}\|z_I\|_{\mathbb{R}^\infty}\|\tilde{\omega}\| + \hbar_{I4}\|z_I\|_{\mathbb{R}^\infty}\end{aligned} \tag{39}$$

where the following symbols are defined respectively, $\hbar_{I1} \triangleq \Upsilon_{a2}\Upsilon_{b4}$, $\hbar_{I2} \triangleq \Upsilon_{a1}\Upsilon_{b4}$, $\hbar_{I3} \triangleq \frac{1}{2}\Upsilon_{b4}\Upsilon_{b5}\Omega_R^{-1}\Omega_g\Omega_{G\eta}$, and $\hbar_{I4} \triangleq \Upsilon_{b4}\Upsilon_{b5}\Omega_\mu + \frac{1}{2}\Upsilon_{b4}\Upsilon_{b5}\Omega_R^{-1}\Omega_g\Omega_{G\zeta_i}$.

From the preceding analysis, substituting (35), (37), (38), and (39) into (33), one derives

$$\begin{aligned}\dot{\mathcal{L}}(t) &\le -\psi_1\hbar_{F1}\|\tilde{\omega}\|^2 - \psi_1\hbar_{F2}\|\tilde{\omega}\| - \psi_1\hbar_{F3}\|z\|^2\|\tilde{\omega}\| + \hbar_{F4}\\ &\times\|z\| + \hbar_{F5}\|\tilde{\omega}\| + \hbar_{F6} - \prod_2\|z_I\|^2 + \hbar_{F7}\|z_I\|_{\mathbb{R}^\infty} - \frac{1}{4}\prod_1\\ &\times\prod_2\|\nabla J^*(\bar{z})\|^2 + (\hbar_{I1} - \frac{\Upsilon_{b3}}{\chi})\|z_I\|_{\mathbb{R}^\infty}^2 + \hbar_{I2}\|z\|\|z_I\|_{\mathbb{R}^\infty}\\ &+ \hbar_{I3}\|z_I\|_{\mathbb{R}^\infty}\|\tilde{\omega}\| + \hbar_{I4}\|z_I\|_{\mathbb{R}^*}\\ &\le -\vartheta_{ii}^T\Theta_{ii}(\chi)\vartheta_{ii} + M_{ii}\vartheta_{ii} + U_a\end{aligned} \tag{40}$$

where $\vartheta_{ii} \triangleq [\|z\|, \|z_I\|_{\mathbb{R}^*}, \|\tilde{\omega}\|]^T$, $M_{ii} \triangleq [\hbar_{F4}, \hbar_{F7} + \hbar_{I4}, -\psi_1\hbar_{F2} + \hbar_{F5}]$, $\Theta_{ii} \triangleq [0, -\frac{1}{2}\hbar_{I2}, 0; -\frac{1}{2}\hbar_{I2}, (\frac{\Upsilon_{b3}}{\chi} - \hbar_{I1}), -\frac{1}{2}\hbar_{I3}; 0, -\frac{1}{2}\hbar_{I3}, \psi_1\hbar_{F1}]$, and $U_a \triangleq \hbar_{F6} - \prod_2\|\bar{z}\|^2 - \frac{1}{4}\prod_1\prod_2\|\nabla J^*(\bar{z})\|^2$.

For matrix $\Theta_{ii}(\chi)$, choosing appropriate parameters can make $\Theta_{ii}(\chi) > 0$. $\dot{\mathcal{L}}(t)$ satisfies $\dot{\mathcal{L}}(t) \le -\underline{\sigma}(\Theta_{ii})\|\vartheta_{ii}\|^2 + \|M_{ii}\|\|\vartheta_{ii}\| + U_a$. If $\|\vartheta_{ii}\| > (\|M_{ii}\| + \sqrt{\|M_{ii}\|^2 + 4\underline{\sigma}(\Theta_{ii})U_a})/(2\underline{\sigma}(\Theta_{ii}))$, then $\dot{\mathcal{L}}(t) \le 0$.

Part Two: $t = t_j$, the self-triggered condition is violated. Meanwhile, the HPDE system (17) updates its self-triggered control strategy and dynamically predict the next triggered moment. Part Two considers only the Lyapunov function difference of (33), which satisfies

$$\begin{aligned}\Delta\mathcal{L}(t) &= \underbrace{J_{\tilde{\omega}}(\tilde{\omega}(t_j)) - J_{\tilde{\omega}}(\tilde{\omega}^-(t_j))}_{\Delta J_{\tilde{\omega}}(\tilde{\omega}(t_j))} + \underbrace{\bar{J}(z(t_j)) - \bar{J}(z^-(t_j))}_{\Delta\bar{J}(z(t_j))}\\ &+ \underbrace{J^*(z(t_{j+1})) - J^*(z(t_j))}_{\Delta J^*(z(t_j))} + \underbrace{J_i(z_l(t_{j+1})) - J_i(z_l(t_j))}_{\Delta J_i(z_l(t_j))}\end{aligned} \tag{41}$$

where $\tilde{\omega}^-(t_j) \triangleq \lim_{\kappa\to 0}\tilde{\omega}(t_j - \kappa)$, $z^-(t_j) \triangleq \lim_{\kappa\to 0} z(t_j - \kappa)$. As can be known from the analysis in Part One, it has $J_{\tilde{\omega}}(\tilde{\omega}(t_j)) \le J_{\tilde{\omega}}(\tilde{\omega}^-(t_j))$ and $\bar{J}(z(t_j) \le \bar{J}(z^-(t_j)$, that is $J_{\tilde{\omega}}(\tilde{\omega}(t_j) \le 0$ and $\Delta\bar{J}(z(t_j) \le 0$. For the last two terms of (41), at the triggered instant $t = t_j$, there exist class-κ functions (Khalil & Grizzle, 2002) such that the following inequalities $\Delta J^*(z(t_j)) \le -\kappa_1(\|z\ (t_{j+1}) - z(t_j)\|)$ and $\Delta J_i(z_l(t_j)) \le -\kappa_1(\|z(t_{j+1}) - z(t_j)\|)$ hold. From this, it is evident that $\Delta\mathcal{L}(t) < 0$ always holds.

In conclusion, through the analysis of the stability of the closed-loop HPDE system (17) under self-triggered control, when $\|\vartheta_{ii}\| > (\|M_{ii}\| + \sqrt{\|M_{ii}\|^2 + 4\underline{\sigma}(\Theta_{ii})U})/(2\underline{\sigma}(\Theta_{ii}))$ is satisfied, $z(t)$, $z_I(t)$ and $\tilde{\omega}(t)$ are all bounded. Since the relationship of $\|Y(s,t)\|_2^2 = \|z(t)\|^2 + \|z_I(t)\|_{\mathbb{R}}^2$ holds in the HPDE system, under the proposed self-triggered ADP optimal control method, the closed-loop HPDE system has a $\mathcal{L}_2$-norm SGUUB. Proof completed. □

In Theorems 1 and 3, we analyzed the finite-dimensional slow and infinite-dimensional fast subsystems, rigorously ensuring overall system stability. Furthermore, Theorem 2 establishes the upper bound of the performance index when ignoring fast dynamics, mathematically guaranteeing that the truncation error is completely bounded and controllable.

Algorithm. Self-Triggered ADP Optimal Control Method for HPDE

HPDE state acquisition: By employing KLD technique to select appropriate spatial basis functions, determine the order N of the low-order slow subsystem, and compute the corresponding temporal modes.
Initialization: Initialize parameters α, ω_1, ω_2, a, b, c. Let the iteration index $i = 0$ and the trigger time $k_t = 0$. ω_2 is the weight at the current time, and ω_1 is the weight at the previous time.
1: **While** $|\|\omega_2\| - \|\omega_1\|| > 10^{-5}$
2: **if** $i == k_t$
3: Update the control using the current weight.
4: Use pdepe to solve the system state at the triggering moment.
5: Predict the next trigger time based on the current trigger time and state.
6: $i = i + 1$
7: **else**
8: Use pdepe to solve the system state at the non-triggering moment.
9: $i = i + 1$
10: **end if**
11: **end while**

For the self-triggered optimal control of spatiotemporal processes, two major hurdles must be overcome compared with existing methods. First, constructing the self-triggered mechanism is exceptionally demanding. Given the inherent nonlinearities and strong spatiotemporal coupling of the underlying HPDE, integrating the optimal control objective with an active prediction mechanism significantly amplifies the design difficulty. Second, establishing closed-loop system stability is highly intricate, as the proof necessitates the simultaneous treatment of flow-jump dynamics and the coupled continuous state evolution. Specifically, one must concurrently analyze the finite-dimensional slow modes and the infinite-dimensional fast modes to guarantee the overall system performance under discrete triggered condition.

Remark 3. To highlight the advantages of the proposed method, we compared its computational complexity with existing approaches. While traditional time-triggered control requires continuous evaluations, and event-triggered control necessitates continuous passive state monitoring, our self-triggered ADP active prediction mechanism calculates the exact next triggering instant based solely on the current state. For instance, if the next update occurs in 10 steps, both time- and event-triggered methods demand 10 updates or evaluations, whereas our method requires only 1 prediction. This completely eliminates continuous monitoring, thereby significantly reducing both computational complexity and hardware dependency.

5. Simulation studies and analysis

In this section, the proposed adaptive self-triggered optimal control method is applied to a catalytic rod reactor to validate its effectiveness and convergence. This system serves as a highly representative model widely adopted by researchers worldwide to verify optimal control methods for HPDE systems.

Nonlinear diffusion-transfer-reaction process in the catalytic rod reactor, which is categorized as a class of HPDE system (Christofides,

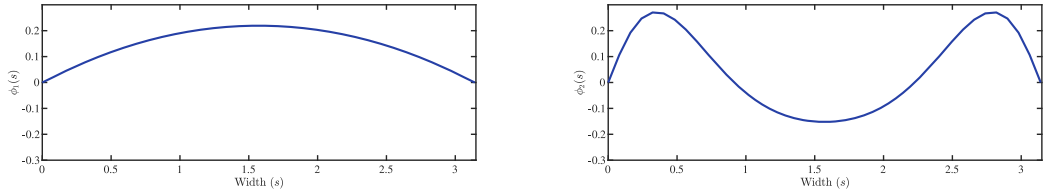


Fig. 1. Spatial basis functions.

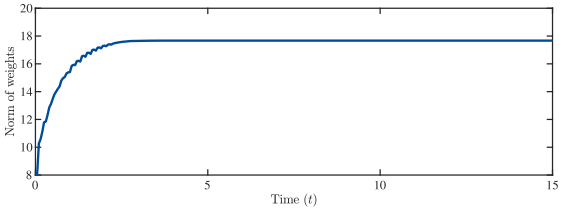


Fig. 2. The convergence process of weights.

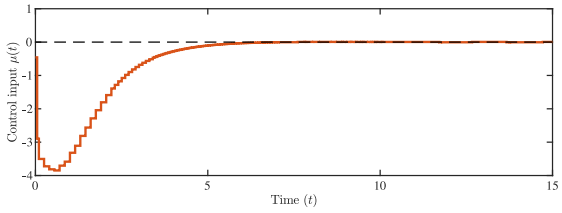


Fig. 3. Self-triggered control strategy.

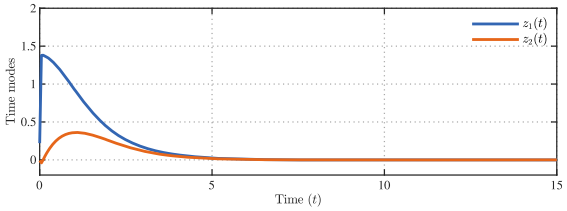


Fig. 4. Low-order slow subsystem time modes.

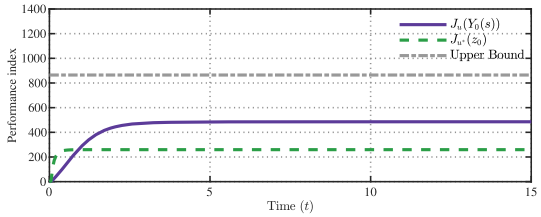


Fig. 5. Performance index.

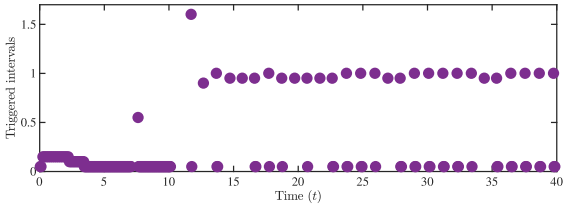


Fig. 6. The distribution of self-triggered intervals.

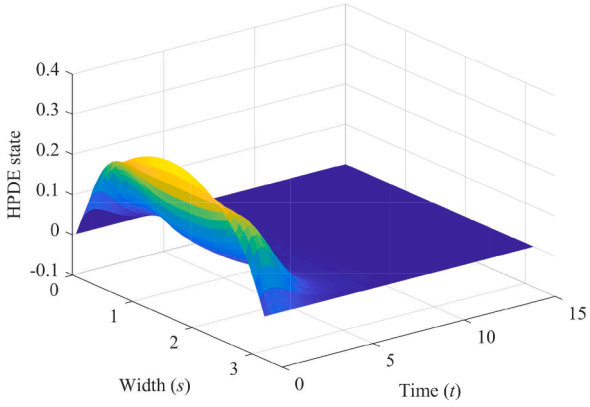


Fig. 7. Evolution process of the HPDE system.

2001), can be described by the following equation,

$$\begin{cases} \dfrac{\partial Y}{\partial t} = k(Y)\dfrac{\partial^2 Y}{\partial s^2} + \eta_1(e^{-\eta_2/(1+y)} - e^{\eta_2}) + \eta_3(\bar{g}(s)u(t) - Y) \\ Y_{out}(t) = \int_0^{\pi} E(s)Y\,ds \end{cases} \tag{42}$$

where Y represents the HPDE state, t is the time variable, and s is the spatial coordinate. $k(Y) = 1$, $\eta_1 = 12$, $\eta_2 = 1$, $\eta_3 = 1$, $g(s) \triangleq H(s - 0.4\pi) - H(s - 0.5\pi)$. The initial condition for the HPDE model in the industrial production process are set as $y_0(s) = 0.3\sin(s)$.

The spatial basis functions, shown in Fig. 1, effectively capture the main dynamic characteristics of the HPDE system and reveal its long-term evolution process. The spatial basis functions selected by the KLD method significantly reduces the computational complexity while retaining the key behavior of the HPDE system.

Next, the learning process for the critic network weights is executed, the parameter in (29) is set as $\psi_1 = 1$. The activation function vector $\eta(z)$ constructed in this section satisfies the relation $\eta(z) \triangleq [z_1^2, z_1 z_2, z_2^2]^T$, and the weight vector $\hat{\omega}$ satisfies the relation $\hat{\omega} \triangleq [\hat{\omega}_1, \hat{\omega}_2, \hat{\omega}_3]^T$. Then, the structure of the critic NN is $\hat{J}(z) = \hat{\omega}_1 z_1^2 + \hat{\omega}_2 z_1 z_2 + \hat{\omega}_3 z_2^2$. During the weight training of the critic NN, the initial value of $\hat{\omega}$ is chosen as $\hat{\omega} = [8, 0, 0]^T$. Fig. 2 illustrates the convergence of the NN weights norm over time, demonstrating the rapid parameter adjustment and optimization during the initial stage. Note that the weight norm eventually stabilizes, indicating that the HPDE system tends to be stable. This convergence process reflects a transition from an initial rapid adjustment phase to a stable convergence phase.

The self-triggered mechanism (18) and (19) makes aperiodic updates of the control strategy, as illustrated in Fig. 3. The self-triggered control strategy $\boldsymbol{\mu}(\bar{z})$ updates only at predicted triggered moments, and remains unchanged otherwise. This mechanism reduces unnecessary control actions and computational overhead while ensuring the closed-loop system stability. Fig. 4 illustrates the evolution of the low-order slow subsystem time modes. From the evolution, the time modes representing the HPDE system dominant dynamics evolve from initial states and asymptotically converge toward zero.

Fig. 5 depicts the performance index convergence of the HPDE system with the self-triggered mechanism, showing rapid convergence to a fixed value with an upper bound specified in Theorem 2.

As shown in Fig. 6, the self-triggered mechanism (18) and (19) contributes to the aperiodic triggered intervals $\varpi(z(t))$, which present a seg-

mented distribution feature. Each point in Fig. 6 is the triggered moment predicted by the self-triggered mechanism. Fig. 7 shows the evolution of the HPDE system states with the ADP self-triggered control. The closed-loop HPDE system gradually converges to near-zero values from its initial state.

The above simulation results demonstrate the convergence of the NN weights, the self-triggered control strategy, and the HPDE system states, while verifying the boundedness of the performance index. Besides, the self-triggered interval sufficiently exhibits the aperiodic distribution characteristics. In conclusion, the simulation results verify the effectiveness and convergence of the proposed method in practical industrial applications.

6. Conclusion

This paper innovatively proposed a self-triggered ADP optimal control method for industrial production processes governed by HPDE system, by introducing the concept of self-triggered mechanism into the HPDE domain. The core of this method lay in designing an appropriate self-triggered mechanism based on the HPDE system stability to dynamically predict the update instants of the control strategy. Through this dynamic prediction, the self-triggered control method fundamentally eliminated the dependence on continuous monitoring of the triggered condition, thereby substantially reducing the huge computational resource consumption. To address the difficulty in obtaining analytical solutions for the HJB equation, this paper employed an ADP based NN critic network to approximate the optimal value function and used an actor network to design the self-triggered optimal control strategy, thereby establishing a novel and adaptive self-triggered control optimization method for the HPDE system. Furthermore, the stability of the self-triggered closed-loop HPDE system and the boundedness of the self-triggered intervals were theoretically analyzed and discussed. Finally, simulation results based on an actual industrial production process demonstrated the effectiveness and convergence of the proposed method.

Although the proposed self-triggered ADP approach lays a solid foundation for the optimal control of HPDE system, several critical issues remain to be explored. Extending this framework to accommodate external disturbances and model uncertainties is of paramount importance for practical industrial applications. Furthermore, incorporating input constraints and actuator saturation represents another pivotal direction for future research.

CRediT authorship contribution statement

Binyan Li: Writing – original draft, Methodology, Conceptualization; **Xiaopeng Cao:** Writing – review & editing, Methodology; **Biao Luo:** Software, Formal analysis; **Shan Xue:** Software, Investigation; **Jiayuan Gao:** Formal analysis, Data curation; **Shaosheng Fan:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.neunet.2026.109853.

Appendix A. Proof of Theorem 1

For the closed-loop HPDE system (17), it is usually necessary to reasonably construct and design the Lyapunov candidate function $J(z,z_I)$ before the system stability analysis. In this section, it is constructed in the form of the sum of $J_i(z_i)$ and $J^*(z)$. The time derivative of $J(z,z_I)$ is given by

$$\begin{aligned}\dot{J}(z,z_I) &= \dot{J}^*(z)+\dot{J}_i(z_I)\\ &= \underbrace{[\nabla J^*(z)]^T(f_i(z,0)+g_i\mu^*(\bar{z}))}_{\mathcal{B}_1}\\ &+ \underbrace{[\nabla J_i(z_I)]^T(A_{I_\chi}z_I/\chi+f_{ii}(z,z_I)+g_{ii}\mu^*(\bar{z}))}_{\mathcal{B}_2}\\ &+ \underbrace{[\nabla J^*(z)]^T(f_i(z,z_I)-f_i(z,0))}_{\mathcal{B}_3}\end{aligned} \tag{A.1}$$

Prior to the stability analysis, through equation transformation and mathematical operations, Eqs. (12) and (13) can be expressed as

$$\begin{cases}[\nabla J^*(z)]^T f_i(z) = -[\nabla J^*(z)]^T g_i u^*(z) - \|z(t)\|_Q^2 - \|u^*(z)\|_R^2\\ g_i^T\nabla J^*(z) = -2Ru^*(z)\end{cases} \tag{A.2}$$

Using Assumptions 1 and 3, substitute (14) and (A.2) into $\mathcal{B}_1$ of (A.1), and the triggered condition is designed and described in (18), the following inequality relationship can be derived,

$$\begin{aligned}\mathcal{B}_1 &= \frac{1}{4}[\nabla J^*(z)]^T g_i R^{-1} g_i^T \nabla J^*(z) - 2[u^*(z)]^T R\mu^*(\bar{z}) - \|z(t)\|_Q^2\\ &= \left\|r^T u^*(z) - r^T\mu^*(\bar{z})\right\|^2 - \left\|r^T\mu^*(\bar{z})\right\|^2 - [(1-\alpha)\bar{z}^T Q\bar{z} + (\frac{1}{\alpha}-1)e^T Qe]\\ &\le \lambda_u^2\|r\|^2\|e(t)\|^2 - \left\|\mu^*(\bar{z})\right\|_R^2 - [(1-\alpha)\|\bar{z}(t)\|_Q^2 + (\frac{1}{\alpha}-1)\|e(t)\|_Q^2]\\ &\le -(1-\alpha)(1-\theta)[\|\bar{z}(t)\|_Q^2 + \left\|\mu^*(\bar{z})\right\|_R^2] - [1-(1-\alpha)(1-\theta)]\left\|\mu^*(\bar{z})\right\|_R^2\\ &- (1-\alpha)\theta\|\bar{z}(t)\|_Q^2 + [\lambda_u^2\|r\|^2 + (\frac{1}{\alpha}-1)\lambda_{\max}(Q)]\|e(t)\|^2\\ &= -(1-\alpha)(1-\theta)[\|\bar{z}(t)\|_Q^2 + \left\|\mu^*(\bar{z})\right\|_R^2] - \Gamma_1\|\bar{z}(t)\|_Q^2 - \Gamma_2\left\|\mu^*(\bar{z})\right\|_R^2\\ &+ [\lambda_u^2\|r\|^2 + \Gamma_3]\|e(t)\|^2\\ &\le -\prod_2\|\bar{z}(t)\|^2 - \frac{1}{4}\prod_1\prod_2\left\|\nabla J^*(\bar{z})\right\|^2\end{aligned} \tag{A.3}$$

where $\prod_1 \triangleq \underline{\sigma}(g_i R^{-1} g_i^T)$, $\underline{\sigma}(\cdot)$ is the lower bound of the singular value of the matrix to which it belongs. $\prod_2 \triangleq (1-\alpha)(1-\theta)$, $\Gamma_1 \triangleq (1-\alpha)\theta$, $\Gamma_2 \triangleq [1-(1-\alpha)(1-\theta)]$, $\Gamma_3 \triangleq (\frac{1}{\alpha}-1)\lambda_{\max}(Q)$.

According to Christofides and Daoutidis (1997), Wang (1964), for the infinite-dimensional system (4), we select a set of positive constants $\Upsilon_{a1}-\Upsilon_{a3}$ and $\Upsilon_{b1}-\Upsilon_{b5}$ based on the reverse Lyapunov theory, the following relationship holds,

$$\begin{cases}\left\|f_{ii}(z,z_I)\right\| \le \Upsilon_{a1}\|z\| + \Upsilon_{a2}\left\|z_I\right\|_{\mathbb{R}^\infty}\\ \left\|f_i(z,z_I) - f_i(z,0)\right\| \le \Upsilon_{a3}\left\|z_I\right\|_{\mathbb{R}^\infty}, for\ \|z\| < \beta_0\\ \Upsilon_{b1}\left\|z_I\right\|_{\mathbb{R}^\infty}^2 \le J_i(z_I) \le \Upsilon_{b2}\left\|z_I\right\|_{\mathbb{R}^\infty}^2\\ \dot{J}_i(z_I) = \frac{1}{\chi}[\nabla J_i]^T A_{I_\chi} z_I \le -\frac{\Upsilon_{b3}}{\chi}\left\|z_I\right\|_{\mathbb{R}^\infty}^2\\ \left\|\nabla J_i(z_I)\right\|_{\mathbb{R}^\infty} \le \Upsilon_{b4}\left\|z_I\right\|_{\mathbb{R}^\infty}\\ \left\|[\nabla J_i]^T g_{ii}\right\| = ([\nabla J_i]^T g_{ii} g_{ii}^T \nabla J_i)^{\frac{1}{2}} \le \Upsilon_{b4}\Upsilon_{b5}\left\|z_I\right\|_{\mathbb{R}^\infty}\end{cases} \tag{A.4}$$

Combining (A.4) and (16), for second sum of (A.1), we can obtain

$$\begin{aligned}\mathcal{B}_2 &= \frac{1}{\chi}[\nabla J_i(z_I)]^T A_{I_\chi} z_I + [\nabla J_i(z_I)]^T f_{ii}(z,z_I) - \frac{1}{2}[\nabla J_i(z_I)]^T g_{ii} R^{-1} g_i^T \nabla J^*(\bar{z})\\ &\le -\frac{\Upsilon_{b3}}{\chi}\left\|z_I\right\|_{\mathbb{R}^\infty}^2 + (\Upsilon_{a1}\|z\| + \Upsilon_{a2}\left\|z_I\right\|_{\mathbb{R}^\infty})\left\|\nabla J_i(z_I)\right\|_{\mathbb{R}^\infty}\end{aligned}$$

$$+\frac{1}{2}\prod_3\left\|[\nabla J_i(z_I)]^T g_{ii}\right\|\|\nabla J^*(\bar{z})\|$$
$$\le \hbar_{I1}\|z_I\|_{\mathbb{R}^\infty}^2+\Upsilon_{a1}\Upsilon_{b4}\|z\|\|z_I\|_{\mathbb{R}^\infty}+\frac{1}{2}\prod_3\Upsilon_{b4}\Upsilon_{b5}\|z_I\|_{\mathbb{R}^\infty}\|\nabla J^*(\bar{z})\| \quad (A.5)$$

where $\prod_3 \triangleq \bar{\sigma}(R^{-1}g_i^T)$, $\bar{\sigma}(\cdot)$ represent the upper bound of the singular value of the corresponding matrix.

Applying the relations from (A.4) to $\mathcal{B}_3$ in (A.1), one has

$$[\nabla J^*(z)]^T(f_i(z,z_I)-f_i(z,0)) \le \Upsilon_{a3}\|z_I\|_{\mathbb{R}^\infty}\|\nabla J^*(z)\| \quad (A.6)$$

The analysis of $\dot{J}(z,z_I)$ shows that $\|\nabla J^*(\bar{z})\|$ and $\|\nabla J^*(z)\|$ coexist, which poses challenges to stability analysis. To address this issue, the following discussion proceeds from two perspectives.

1) When $\|\nabla J^*(\bar{z})\| \ge \|\nabla J^*(z)\|$, $\Upsilon_{a3}\|z_I\|_{\mathbb{R}^\infty}\|\nabla J^*(z)\| \le \Upsilon_{a3}\|z_I\|_{\mathbb{R}^\infty}\|\nabla J^*(\bar{z})\|$ can be obtained. Using (A.3), (A.5), and (A.6) for (A.1), the result is that $\dot{J}(z,z_I)$ satisfies the following relationship,

$$\dot{J}(z,z_I) \le -\vartheta_i^T\Theta_i(\chi)\vartheta_i + U_b \quad (A.7)$$

where $\vartheta_i \triangleq [\|z\|,\|z_I\|_{\mathbb{R}^\infty},\|\nabla J^*(\bar{z})\|]^T$, $\Theta_i \triangleq [0,-\frac{1}{2}\Upsilon_{a1}\Upsilon_{b4},0;-\frac{1}{2}\Upsilon_{a1}\Upsilon_{b4},\hbar_{I1},-\frac{1}{2}\Upsilon_{a3}-\frac{1}{4}\prod_3\Upsilon_{b4}\Upsilon_{b5};0,-\frac{1}{2}\Upsilon_{a3}-\frac{1}{4}\prod_3\Upsilon_{b4}\Upsilon_{b5},\frac{\prod_1\prod_2}{4}]$, and $U_b \triangleq -\prod_2\|\bar{z}\|^2$. If appropriate parameters are chosen such that $\Theta_i(\chi)>0$, then $-\vartheta_i^T\Theta_i(\chi)\vartheta_i \le 0$ can be derived, and we have $\dot{J}(z,z_I)\le 0$.

2) Conversely, it holds $\|\nabla J^*(\bar{z})\| < \|\nabla J^*(z)\|$. According to Assumptions 3, $\dot{J}(z,z_I)$ can be transformed into

$$\dot{J}(z,z_I) \le -\vartheta_i^T\Theta_i(\chi)\vartheta_i + M_i\vartheta_i \quad (A.8)$$

where $M_i \triangleq [0,\Upsilon_{a3}\lambda_V\|e\|,0]$. Given that $\Theta_i(\chi)>0$, (A.8) becomes $\dot{J}(z,z_I) \le -\underline{\sigma}(\Theta_i)\|\vartheta_i\|^2 + \|M_i\|\|\vartheta_i\|$. If $\|\vartheta_i\| > \vartheta_M$ and $\vartheta_M \triangleq \|M_i\|/\underline{\sigma}(\Theta_i)$, and then system (4) is SGUUB.

Considering the closed-loop HPDE system (17), $\|Y(s,t)\|_2^2 = \|z(t)\|^2 + \|z_I(t)\|_{\mathbb{R}^\infty}^2$ is satisfied. Then, under the effect of the triggered condition (18), the closed-loop HPDE system is an SGUUB with the $\mathcal{L}_2$-norm.

On this basis, the self-triggered control moments of the HPDE system is predicted and calculated. From Assumption 1, as for the self-triggered control strategy (16), $\mu^*(\bar{z})$ can be derived from the following relationship,

$$\|\mu^*(\bar{z})\| \le \|u^*(z)-\mu^*(\bar{z})\| + \|u^*(z)\|$$
$$\le \lambda_u\|e(t)\| + \|u^*(z)\| \quad (A.9)$$

Furthermore, considering that the low-order slow subsystem of the HPDE system can be described in the form shown in (6), we have

$$\left\|\left\langle \dot{Y}(s,t),\phi_F(s)\right\rangle\right\| = \|\dot{z}\| \le \|f_i(z(t))\| + \|g_i\mu^*(\bar{z})\|$$
$$\le \lambda_f\|z\| + \bar{\sigma}(g_i)\|\mu^*(\bar{z})\| \quad (A.10)$$

By substituting (A.9) into (A.10), one obtains

$$\|\dot{z}\| \le \lambda_f\|z\| + \lambda_u\bar{\sigma}(g_i)\|e(t)\| + \bar{\sigma}(g_i)\|u^*(z)\| \quad (A.11)$$

In combination with (31), it is further found that (A.11) can be transformed,

$$\|\dot{z}\| \le \lambda_f\|z\| + \lambda_u\bar{\sigma}(g_i)\|e(t)\| + \bar{\sigma}(g_i)\Omega_\mu \quad (A.12)$$

Based on (A.13), let $a=\lambda_f$, $b=\lambda_u\bar{\sigma}(g_i)$, $c=\bar{\sigma}(g_i)\Omega_\mu$, and $d=a+b$, it yields

$$\|\dot{z}\| \le a\|z\| + b\|e(t)\| + c \quad (A.13)$$

For each moment $t\in[t_j,t_{j+1})$ when the condition (18) is satisfied, from (A.13) and (14), we obtain

$$\|\dot{e}(t)\| = \|\dot{z}\| \le a\|\bar{z}\| + d\|e(t)\| + c \quad (A.14)$$

Further, by combining (A.14) with the comparison principle, it delivers

$$\|e(t)\| \le \frac{a\|\bar{z}\|+c}{d}(e^{d(t-t_j)}-1) \quad (A.15)$$

At the moment when the self-triggered condition is violated, i.e., $t=t_{j+1}$, we have

$$\frac{a\|\bar{z}\|+c}{d}(e^{d(t_{j+1}-t_j)}-1) \ge \left\|e(t_{j+1})\right\| > e_T \quad (A.16)$$

By combining the inequality transformation, the two adjacent self-triggered moments satisfy the following relation,

$$t_{j+1}-t_j > \frac{1}{d}\ln\left(1+\left(\frac{d}{a\|\bar{z}\|+c}e_T\right)\right) > 0 \quad (A.17)$$

Subsequently, according to (18) and (A.17), the next self-triggered moment t_{j+1} can be predicted as

$$t_{j+1} \triangleq t_j + \frac{1}{d}\ln\left(1+\left(\frac{d}{a\|\bar{z}\|+c}\times\sqrt{\frac{\Gamma_1\|\bar{z}\|_Q^2+\Gamma_2\|\mu^*(\bar{z})\|^2}{\lambda_u^2\|r\|^2+\Gamma_3}}\right)\right) \quad (A.18)$$

Given the designed self-triggered mechanism (18) and (19), the closed-loop HPDE system (17) is guaranteed to be SGUUB under self-triggered control. And the proof is concluded. □

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Binyan Li received the B.S. degree in automation and the M.S. degree in electrical engineering from the School of Information Science, Xiangtan University, Xiangtan, China, in 2014 and 2017, respectively, and the Ph.D. degree from the School of Automation, Institute of Control Engineering, Central South University, Changsha, China, in 2022. She is currently a Lecturer with the School of Electrical & Information Engineering, Changsha University of Science & Technology, Changsha. She is a member of CAA and CAAI. Her interests are in modeling and optimal control of distributed parameter systems, such as temperature field in roller kiln, event-triggered optimal control, and adaptive dynamic programming.



Xiaopeng Cao received the B.S. degree in the School of Electrical and Control Engineering, Henan University of Urban Construction, China, in 2024 and began to study postgraduate program in the School of Artificial Intelligence, Changsha University of Science and Technology, Changsha, China. He current research direction is the modeling and optimization control methods of industrial process temperature fields.



Biao Luo received the Ph.D. degree in control science and engineering from Beihang University, Beijing, China, in 2014.He is currently a Professor with the School of Automation, Central South University (CSU), Changsha, China. Before joining CSU, he was an Associate Professor and Assistant Professor with the Institute of Automation, Chinese Academy of Sciences, Beijing, China, from 2014 to 2018. He serves as an Associate Editor for the IEEE Transactions on Neural Networks and Learning Systems, the Artificial Intelligence Review, the Neurocomputing, and the Journal of Industrial and Management Optimization. He is a Senior Member of the IEEE, and the Vice-Chair of Adaptive Dynamic Programming and Reinforcement Learning Technical Committee, Chinese Association of Automation. His current research interests include intelligent control, reinforcement learning, deep learning, and decision-making.



Shan Xue received the Ph.D. degree from the School of Computer Science and Engineering, South China University of Technology, in 2023. She has authored and co-authored over 20 journal and conference papers. Her papers have been cited more than 1000 times according to Google Scholar. She is a member of IEEE, CAA, CAAI, and CICC. She currently serves as an Associate Editor for the SCI journals IEEE Transactions on Systems, Man, and Cybernetics: Systems and Artificial Intelligence Review. Her current research interests include intelligent control, adaptive dynamic programming, reinforcement learning for control, event-triggering mechanism, control and modeling of complex nonlinear systems, and neural networks.



Jiayuan Gao received the B.S. degree in Electrical Engineering from the Henan Polytechnic University, Henan, China, in 2015, the M.S. degree in Shanghai University of Electric Power, Shanghai, China, in 2018, and the Ph.D. degree in electrical engineering from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2022. Now, he has been working as a lecturer at the Changsha University of Science & Technology. His research interests include grid-connected converter control and the stability control of renewable energy power generation systems.



Shaosheng Fan received the B.S. degree in electrical engineering from Southwest Jiao Tong University, Chengdu, China, in 1987, and the M.S. and Ph.D. degrees in control engineering from Hunan University, Changsha, China, in 1996 and 2005, respectively. From 1987 to 1993, he was an Electrical Engineer with China South Locomotive & Rolling Stock Corporation Limited, Beijing, China. Since 1996, he has been with the School of Electrical and Information Engineering, Changsha University of Science & Technology, Changsha, where he is currently a Professor. His research interests include image processing, deep learning, and electric robot.